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14 August 1980
SUPERSEDING
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18 October 1974

MILITARY SPECIFICATION

PIPE, STEEL, WEIGHT-COATED, FOR SUBMARINE PIPELINE SYSTEMS

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers weight-coated steel pipe for use with submarine pipeline systems.

1.2 Classification. The pipe shall be the following sizes, as specified (see 6.2):

12 - inch
16 - inch
20 - inch

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 4710

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-775	- Steel Sheets, Carbon, Zinc-Coated (Galvanized) by the Hot-Dip Process.
QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-D-723	- Drums, Fiber.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129	- Marking For Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN CONCRETE INSTITUTE (ACI)

ACI 306 - Recommended Practice For Cold Weather Concreting.

(Application for copies should be addressed to the American Concrete Institute, 22400 West Seven Mile Road, Detroit, MI 48219.)

AMERICAN PETROLEUM INSTITUTE (API)

SPEC-5L - Line Pipe.

(Application for copies should be addressed to the American Petroleum Institute, 2101 L Street, NW, Washington, DC 20037.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 615 - Deformed and Plain Billet-Steel Bars For Concrete Reinforcement.
C 29 - Unit Weight and Voids in Aggregate.
C 39 - Compressive Strength of Cylindrical Concrete Specimens.
C 150 - Portland Cement.
C 309 - Liquid Membrane-Forming Compounds For Curing Concrete.
D 1557 - Moisture-Density Relations of Soils, Using 10-lb (4.5 kg) Rammer and 18-inch (457 mm) Drop.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C203 - Coal-Tar Protective Coatings and Linings For Steel Water Pipelines--Enamel and Tape--Hot-applied.

(Application for copies should be addressed to the American Water Works Association, Inc., 6666 West Quincy Avenue, Denver, CO 80235.)

ASPHALT INSTITUTE

SS-7 - Asphalt Protective Coatings For Pipe Lines.

(Application for copies should be addressed to the Asphalt Institute, Asphalt Institute Building, College Park, MD 20740.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The coated pipe shall consist of lengths of steel pipe on the exterior of which is placed a corrosion protection coating and an outer protective, weighted coating.

3.2 First article (first produced pipes). The contractor shall furnish one or more pipes for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce pipes that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Steel pipe. The pipe shall conform to API-SPEC-5L, Grade B, plain beveled ends, and shall be furnished in 35- to 50-foot lengths. Wall thickness and weight of the pipe shall be as follows:

Nominal diameter (inches)	Wall thickness (inch)	Weight/linear foot (pounds)
12	0.250	33.38
16	0.312	52.27
20	0.375	78.60

3.4.1 Pipe preparation. Each length of pipe shall be cleaned of all oil and grease. Rust, mill scale, or other deleterious substances shall be removed by means of blast type cleaning equipment. Cleaning shall not be permitted on wet or moist pipes.

3.5 Coatings.

3.5.1 Undercoat. The undercoating shall consist of either of the following at the contractor's option:

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(a) Coal-tar primer and enamel with spirally-wrapped fiberglass, and then wrapped with No. 15 tar-saturated asbestos felt as specified in 3.5.1.1, or
(b) An asphalt-mastic type coating as specified in 3.5.1.2.

3.5.1.1 Coal primer and enamel.

3.5.1.1.1 Primer. The primer for enamel-type coating shall conform to AWWA C203, Type A (coal-tar) primer.

3.5.1.1.2 Enamel. Enamel shall conform to AWWA C203, cold temperature enamel.

3.5.1.1.3 Wrap. The wrap shall be flexible glass fiber with physical properties not less than the following:

(a) Tensile strength (longitudinal)	45 pounds per 3-inch width (min.)
(b) Thickness	0.017 inch to 0.023 inch
(c) Weight	1.1 pound per 100 square feet (nominal)

3.5.1.1.4 Tar-saturated asbestos. Tar-saturated asbestos wrap shall conform to AWWA C203.

3.5.1.1.5 Application.

3.5.1.1.5.1 Priming. Immediately after the pipe is cleaned, the primer shall be applied to the pipe to obtain complete coverage with no holidays or runs or globules of primer. The primer shall not be applied over wet, moist, or frosty pipe, and freshly primed pipe shall not be allowed to come into contact with earth, leaves, or other foreign matter.

3.5.1.1.5.2 Preparation of enamel. The enamel shall be prepared in a sufficient number of kettles so that the enamel is uniformly heated to correct temperature prior to application on the pipe. Accurate thermometers shall be furnished and maintained for each kettle for determining the temperature to which the enamel is heated. Enamel shall be stirred during the heating and application of the charge. Kettles

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shall not be permitted to act as continuous enamel supply sources by adding unmelted enamel during the time the kettles are in use, but shall be emptied of one charge and cleaned before the next charge of unmelted enamel is added; except, when mechanically agitated kettles are used (continuous agitation), the kettle need not be completely emptied before recharging. Enamel shall be condemned and discarded as unfit for use when it has become damaged by overheating or by continued heating.

3.5.1.1.5.3 Preparation of primed pipe for enamel. After the priming coat has become completely dry and immediately before the application of the enamel, the surface of the primed pipe shall be thoroughly cleaned and free of all dirt, dust, or any foreign matter to insure the maximum strength of bond between the enamel and the pipe. Clean rags, not burlap bags, shall be used in the dusting or cleaning of the primed pipe prior to the application of the enamel.

3.5.1.1.5.4 Wrapping. The inner and outer wrappings and the enamel shall be applied to the cleaned, primed pipe in the following sequence:

(a) Apply enamel to a thickness of 0.0937 inch plus or minus 0.0312 inch.
(b) Spirally apply one layer of fiberglass mat with an overlap of not less than 1 inch.
(c) Apply enamel to a minimum thickness of 0.0625 inch.
(d) Spirally apply one layer of coal-tar-saturated asbestos felt, completely bonded to the enamel with an overlap of not less than 1 inch. (Total thickness, including both wraps, shall be approximately 0.1875 inch.)

3.5.1.2 Asphalt-mastic coating.

3.5.1.2.1 Primer. Primer for the asphalt-mastic undercoat shall conform to Asphalt Institute SS-7, mastic systems.

3.5.1.2.2 Mastic. Materials for the mastic shall be the following proportions:

Material	Percent by weight
Asphalt	10-12
Aggregate	88-90
Fiberglass	0.10-0.20

All materials shall be heated, metered in the above proportions and completely mixed.

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3.5.1.2.3 Asphalt. Asphalt for asphalt-mastic-type undercoating shall be petroleum asphalt containing no mineral matter other than that naturally present in the asphalt; shall be uniform in character and free from water; shall not foam when heated to 400° F; and shall conform to the following:

(a)	Operating temperature, (F)	125°
(b)	Softening point (ring and ball), (F)	175°-200°
(c)	Penetration (at 77° F, 100g, 5 sec), (tenths of mm)	15-17
(d)	Flash point (Cleveland open cup), (F)	450°+
(e)	Loss on heating at 325° F, 5 hr, (percent)	0.5-
(f)	Ductility at 77° F, (cm)	2.5+
(g)	Percent soluble in CCl_4	99.0+

3.5.1.2.4 Aggregate. Aggregate for asphalt-mastic-type undercoating shall be a clean, nonmicaceous mixture of sand and mineral filler, free from organic impurities and other deleterious substances, and shall be as follows.

3.5.1.2.4.1 Sand. The sand shall conform to the following requirements when graded from coarse to fine by means of a standard Tyler screen:

Sieve size number	Passing, percent by weight
6	100
16	52-66
50	31-37
100	25-30

3.5.1.2.4.2 Mineral filler. The mineral filler shall be pulverized limestone or equivalent portion of the aggregate that shall be added until 20 to 26 percent of the total aggregate passes the 200-mesh standard Tyler screen.

3.5.1.2.5 Fiberglass. Fiberglass length shall be predominantly 0.250 to one inch with nominal diameter of 0.0001 to 0.001 inch.

3.5.1.2.6 Application.

3.5.1.2.6.1 Primer. The primer shall be applied as specified in 3.5.1.1.5.1.

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3.5.1.2.6.2 Mastic application. The hot mastic shall be applied to the dry, primed surface of the pipe either by extrusion or by hand troweling to form a continuous and seamless coating thoroughly bonded to the pipe. All holidays and damaged areas shall be repaired using the same materials and at the same temperature as the original coating. The thickness of the mastic undercoating shall be as follows:

Pipe size, inches	Mastic thickness	
	Minimum inch	Desired inch
12	0.375	0.5
16	0.4375	0.5625
20	0.5	0.625

3.5.1.2.6.3 Cleaning after undercoating. Each end of each length of pipe shall be cleaned and free of all traces of undercoating for a distance of 9 inches from the beveled end of the pipe. Coating material shall not be allowed to remain on the inside surface of the pipe. The end of the asphalt-mastic coating shall be tapered 1 in 4.

3.5.1.3 Storing after undercoating. After undercoating, the length of pipe shall be stored as follows:

(a) During inspection and prior to being concrete coated, the enamel-wrapped pipe shall be stored on canted boards or on other timber structures so that the entire weight of the joint shall be supported by the bare ends of the joint.

(b) Asphalt-mastic-type undercoating shall be allowed to cool to an extent necessary to support its own weight. The pipe shall be stored on soft earth berms of sufficient height to support the length of pipe clear of the ground, or shall be supported over its entire length on a smooth earth or sand surface which is free from lumps, rocks, or other material which might chip, fracture, dent, or otherwise damage the undercoating.

3.5.2 Outer coating. The outer coating shall be concrete consisting of Portland cement conforming to ASTM C 150, aggregate, and water. When applied as specified herein, with reinforcing, the concrete shall have a

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compressive strength of not less than 2,000 pounds per square inch (psi) in 24 hours, and 3,500 psi in 7 days when tested as specified in 4.5.2.2. The concrete shall have a density of 190 pounds per cubic foot (lb/cu ft), plus or minus 5 pounds.

3.5.2.1 Aggregate. Concrete aggregate shall be clean and free from injurious amounts of clay, salt, alkali, or organic impurities that may affect the strength of the concrete. The aggregate shall be a high silica content, low grade iron ore (taconite, hematite, magnetite, or ilmenite). The aggregate shall have a specific gravity or not less than 4.50, and hardness of 5 to 6.50 (Mohs scale). The aggregate shall conform to the following when graded from coarse to fine by means of standard sieves:

Sieve size	Percent passing
0.25-inch	100
No. 16	30-75
No. 50	10-40
No. 100	0-15

Unit weight (dry, rodded bulk density) shall be not less than 175 lb/cu ft as determined by ASTM C 29.

3.5.2.2 Water. Mixing water shall be free from deleterious substances harmful to the concrete coating.

3.5.2.3 Reinforcing wire fabric. Reinforcing wire fabric shall be 1.50-inch mesh formed from 17-gage steel wire conforming to ASTM A 615. The wire shall be galvanized after being formed into the fabric.

3.5.2.4 Curing compound. Curing compound shall be the clear membrane type conforming to ASTM C 309, Type 1.

3.5.2.5 Negative buoyancy. Pipe of the sizes and weights listed below shall have the following negative buoyancies after coating with the concrete:

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Pipe size (inches)	Negative buoyancy (lb/linear ft)
12	19
16	30
20	32

The calculations to determine the coating thickness necessary to provide the negative buoyancy required shall be based on the following:

(a) Density of concrete - 190 lb/cu ft plus or minus 5 lb.
(b) 64 lb/cu ft sea water.
(c) 12-inch cutback with no "field joints" to be applied. The calculated thicknesses of the coating in the coated portion of the pipe shall compensate for the uncoated portion. (Military contingencies may necessitate installation of submerged lines without application of field coating at joints.)
(d) Pipe filled with air during installation of submarine pipeline.

3.5.2.6 Mixing. The dry materials shall be mixed to a homogeneous mixture before the addition of water.

3.5.2.7 Application. After the addition of water to the dry mixture, the concrete shall be applied by a high-speed applicator. This mixture shall then be discharged from the machine through high-speed belts or brushes or a combination of both so that an impact velocity of 4,000 to 5,000 feet per minute is obtained. The pipe shall be rotated and passed through the discharge stream with a speed of rotation and longitudinal velocity so that the specified thickness of concrete coat is obtained. Irregularities in the finished concrete surface shall be not more than 0.5 inch. Simultaneous with the application of concrete, reinforcing wire fabric shall be fed onto the pipe to provide a continuous layer of reinforcement approximately midway between concrete coatings. The number of reinforcing wraps for the various thicknesses of concrete coatings shall be as follows:

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Thickness of coating (inches)	Number of wraps
1 - 1.75	One wrap
1.875 - 3.25	Two wraps
3.375 - 4.50	Three wraps
4.50 and over	Four wraps

To insure continuity of reinforcement, an overlap of not less than 0.5 inch shall be maintained. Reinforcing wire fabric shall be extended from the concrete toward the bevel for a distance of 2 inches plus or minus 0.50 inch, so that field joint reinforcing wire fabric may be attached to provide continuity of reinforcement through the field joint.

3.5.2.7.1 Low-temperature application. The outer coat shall be applied when the surrounding temperature is 40° F or higher. During the curing period, whenever the temperature drops below 40° F, the outer coat shall be protected in accordance with ACI 306.

3.5.2.8 Cutback. The concrete outer coat shall not be applied to within 12 inches plus or minus 1 inch from each end of the length of pipe. Grout or concrete shall not be allowed to remain on the outside of the cutback or inside of the pipe.

3.5.2.8.1 Curing-compound application. The curing compound shall be applied immediately after the concrete outer coat is applied. The application shall conform to the recommendations of the manufacturer of the curing compound.

3.6 Weighing. Each length of pipe shall be weighed after application of the concrete curing compound. The weight of any individual length of pipe shall not vary by more than plus or minus 5 percent from that specified in 3.5.2.5 for negative buoyancy.

3.7 Patching. The outer coating shall be patched as specified herein. Coating found to be defective at the time of application shall be repaired before the pipe is removed from the coating apparatus and shall be cured in accordance with 3.11. Patches shall conform to 3.5.2 and 3.5.2.7. Reinforcing wire fabric shall be restored whenever necessary. When reinforcing wire fabric is restored, the continuity of reinforcement shall be maintained.

3.7.1 Patching materials. Patching materials shall be the same as those from which the overcoating is made, so that the resulting coating shall be equal in weight, density, uniformity, strength, and characteristics to the originally applied overcoating.

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3.8 Spalling. Spalling of more than 20 percent of the total thickness of the overcoating shall be repaired. An area of less than 1 square foot where the remaining concrete is sound need not be repaired unless the reinforcing wire fabric is exposed. An area of more than 1 square foot and less than 3 square feet shall be prepared for patching by exposing the reinforcing wire fabric throughout the entire damaged area and by undercutting the edges of the sound concrete to provide a key lock. An area of more than 3 square feet and less than 1 foot long shall be prepared for patching by removing the coating around the entire periphery of the pipe throughout the entire damaged area, and an area of more than 1 foot long shall be repaired on the coating machine.

3.9 Cracking. Concrete coating cracks shall be repaired when the cracks are more than 0.0625 inch wide and extend for more than half the circumference of the pipe, or when cracks are over 12 inches in length longitudinally along the pipe, irrespective of width. Preparation for patching shall be made by chiseling the crack out to a width not to exceed 1 inch throughout the length of the crack.

3.10 Reweighing. Lengths of pipe patched on the machine shall be reweighed and shall conform to 3.6.

3.11 Curing.

3.11.1 Curing of lengths of pipe. Each length of freshly concrete-coated pipe shall be stored in or on sand piles or windrows of sand for a period of not less than 4 days.

3.11.1.1 Curing of patches. Lengths of pipe that have been damaged after curing shall be repaired, and shall be cured for not less than 36 hours prior to any further handling.

3.12 Stockpiling.

3.12.1 Uncoated steel pipe. Uncoated steel pipe shall be stored not less than 1 foot above the natural grade of the ground.

3.12.1.2 Coated steel pipe. Coated steel pipe shall be racked not more than three layers high after the curing period. Lengths of coated pipe shall not be racked until they have been inspected, cleaned on the bare ends, and patched when required.

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3.13 Bevels. The bevel on the ends of the pipe shall not be dented, scarred, mutilated, or otherwise damaged during handling.

3.14 Field joint-coating and joint-forming materials. The contractor shall furnish all materials required for field coating the joints, such as reinforcing wire fabric, tape, tape adhesive, forms, banding steel, and bagged, dry, concrete mix. Materials shall be furnished in the quantities and sizes to correspond to the size pipe and the number of field joints required, based on the amount of pipe and joints covered by the contract or purchase order, plus a 10 percent waste.

3.14.1 Tape adhesive (primer). Tape adhesive for field joints shall be of the polyvinyl chloride type.

3.14.2 Tape. Tape for field joints shall be the polyvinyl chloride type. Width of the tape shall be suitable for the size of the pipe specified herein.

3.14.3 Concrete for field application. The dry mix for field application shall be as specified in 3.5.2 and 3.5.2.1. The contractor shall determine the amount of additive cement required to be added to the mix due to the increased amount of water used in the pouring of a field joint. The contractor shall include instructions specifying the amount of water required per bag of mix for regular pouring in the field so as to maintain approximately the same water-to-cement ratio as that used in the coating of the pipe.

3.14.4 Reinforcing wire fabric. Reinforcing wire fabric shall be as specified in 3.5.2.3 but shall be furnished in 24-inch-wide rolls of a standard length.

3.14.5 Sheet metal. The sheet metal used to form the coating on a field joint shall conform to QQ-S-775, 28 gage, 50 inches long and of a width to enclose the concrete-coated pipe.

3.14.6 Banding steel and clips. Banding steel or straps for securing the form to the pipe shall conform to QQ-S-781, 1.25 inches wide and 0.035 inch thick. Three straps and three clips per joint shall be furnished.

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3.15 Marking.

3.15.1 Identification marking. The coated pipe and all other supplies shall be identified in accordance with MIL-STD-130 and as specified herein.

3.15.2 Internal marking. The following data shall be legibly applied with white paint in 2-inch numerals inside each joint of pipe at least 1 inch from each end:

(a) Length of joint in feet and inches to nearest inch.
(b) Total weight of joint to nearest 50 pounds.

The data shall be as shown in the following example:

Data	Example
Length, 41 ft-0.75 inch	41-1
Weight, 4,340 pounds	4350

3.16 Workmanship. The finished pipe shall be clean and shall be free from dirt, dents, cracks or other imperfections. The ends of the pipe shall be clean with no rust, dents, debris, or deposits of the coating compounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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(a) First produced pipe inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced pipe inspection.

4.3.1 Examination. The first produced pipe shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced pipe shall be tested as specified in 4.5.2.1 through 4.5.2.4. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Each length of pipe shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples of pipe and coating material selected in accordance with 4.4.1 shall be tested as specified 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination.

4.5.1.1 In-process examination. During application of each layer of protective coating, the pipe shall be examined for the following defects:

Major

101. Pipe not cleaned as specified before coating application.
102. Holidays or imperfections in the coating.
103. Coatings not continuous.
104. Coatings not smooth.
105. Coating thickness not as specified.
106. Reinforcing wire fabric not imbedded approximately midway in concrete coating.
107. Reinforcing wire fabric not overlapped as specified.
108. Curing compound not applied as specified.

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109. Curing time and method of curing not as specified.
110. Preparation for patching and finished patch not as specified.
111. Presence of cracks at junction of patch with coating.

4.5.1.2 Final examination. The finished pipe shall be examined for the following defects:

112. Material not as specified.
113. Concrete coating with irregularities of more than 0.375 inch above or below the general coating level.
114. Cutback of either coating not as specified.
115. Reinforcing wire fabric not extended from concrete toward bevel as specified.
116. Uncoated surface of pipe at cutback and inside of pipe not clear of all adhesive tape and concrete as specified.
117. Stockpiling not as specified.
118. Field-joint material not as specified.
119. Field-joint quantities not as required or specified.
120. Markings missing or not as specified.
121. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Asphalt-mastic undercoating. When applicable, prior to application of the concrete overcoating, the asphalt-mastic undercoating shall be tested with a high-voltage electric holiday detector. Nonconformance to 3.5.1.2.6.2 shall constitute failure of this test.

4.5.2.2 Compressive strength. The concrete shall be subjected to the tests specified in ASTM C 39 on samples of the material taken while the material is being applied to the pipe; the sample shall be taken at the beginning and end of each day's operation, and shall be molded into 0.03 cubic-foot cylinders as specified in ASTM D 1557. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.3 Concrete coating application. The impact velocity of the coating on the pipe shall be determined by calculations based on the measured velocity or revolutions per minute of the belt or brushes. Nonconformance to 3.5.2.7 shall constitute failure of this test.

4.5.2.4 Weight. Each length of pipe shall be weighed after application of the curing compound as specified in 3.5.2.8.1. Nonconformance to 3.6 shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.1.1 Unit of product. For the purpose of inspection, a length of pipe with accessories prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

122.	Materials, methods, and containers not as specified for Level A or Level B. Each incorrect material, method, or container shall be considered one defect.
123.	Polyethylene bag containing cement-aggregate dry mix not sealed for Level A or Level B.
124.	Strapping not as specified for Level A or Level B.
125.	Drums not palletized as specified for Level A or Level B.
126.	Marking missing, illegible, incorrect, or incomplete for Level A, Level B or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified in 6.2.

5.1.1 Level A.

5.1.1.1 Field joint-coating and joint-forming materials.

5.1.1.1.1 Tape. Tape required for a complete installation shall be placed in boxes conforming to PPP-B-566, Variety 2, PPP-B-676, Variety 2, or PPP-B-636 Class Weather-Resistant.

5.1.1.1.2 Adhesive. Adhesive required for a complete installation shall be preserved as specified for tape in 5.1.1.

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5.1.1.1.3 Banding steel. The required banding steel shall be preserved in accordance with the Level A packaging requirements of QQ-S-781.

5.1.1.1.4 Clips. Clips required for a complete installation shall be placed in a fiberboard box conforming to PPP-B-636, Class Weather-Resistant.

5.1.2 Commercial. The material required for field-joining the pipes shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B or Commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pipe sections. The pipe sections shall not require packing.

5.2.1.2 Concrete coating mix. The cement and aggregate mix required for one complete installation shall be packed in fiber drums conforming to PPP-D-723, Type III, Grade A, 300-pound capacity. Drums shall be 15.50 inches in diameter by 14.75 inches in height (inside dimensions) and shall have a polyethylene liner of 6 mil thickness conforming to L-P-378. The liner shall be heat sealed closed after filling. The drums shall be palletized in accordance with MIL-STD-147, load Type XIII. The pallet load shall have a canvas cover (storage aid 8) or a waterproof wrap (storage aid 9).

5.2.1.3 Other items. Other items required for the field-joining of the pipe sections, in quantities required for one complete installation, shall be packed in a minimum number of boxes conforming to PPP-B-601, Overseas Type, or PPP-B-621, Class 2. Closure and strapping of the boxes shall be in accordance with the appendix to the applicable box specification.

5.2.2 Level B packing. Level B packing shall be the same as for Level A in 5.2.1 except boxes conforming to PPP-B-601 or PPP-B-621 shall be Domestic Type or class as applicable.

5.2.3 Commercial. The weight-coated pipe and required field joint-coating and joint-forming materials shall be packed in accordance with MIL-STD-1188.

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5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The coated pipe is intended primarily for transporting petroleum products in submarine pipelines for offshore tanker loading and unloading facilities, or for underwater pipeline river crossings.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size of pipe required (see 1.2).
(c) The time frame required for submission of first produced pipe and number of pipes required (see 3.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced pipe. Any changes or deviations of production coated pipe from the first produced coated pipe during production will be subject to the approval of the contracting officer. Approval of the first produced coated pipe will not relieve the contractor of his obligation to furnish coated pipe conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pipe.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of pipes.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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